A Minor Report on,

Electronic Toll Collection System

Under Subject of

Design Engineering

B.E , Semester - VI

Prepared By,

Dhruval Shah (140280117010)
Parth Panchal (140280117033)
Darshil Shah (140280117050)
Mittal Vegda (140280117061)

Under the Guidance of
Prof. **Nidhi Kanani**

Department of Instrumrntation and Control enginerring
LD College of Engineering, Ahmedabad.

CERTIFICATE

This is to certify that the Minor Project Report entitled **"Electronic Toll Collection System"** submitted by Dhruval Shah (140280117010),Parth Panchal (140280117033),Darshil Shah (140280117050),Mittal Vegda (140280117061) as the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology inInstrument and Control Department, LD College of Engineering is the record of work carried out by them under my supervision and guidance. The work submitted in our opinion has reached a level required for being accepted for the examination.

Date:

Prof. Nidhi Kanani **Prof. M.J.Modi**

Project Guide **HOD (IC)**

ACKNOWLEDGEMENT

We owe a great thanks to many people who helped and supported us during the completion of this project. Our deepest thanks go to **Mrs.Nidhi Kanani**, Assistant Professor, Department of Instrumentation and Control Engineering who acted as our mentor and guided us throughout this project. We thank her for giving ideas and briefs for the work that had to be done. A special thanks to **Mr. M.J.Modi**, Head of Department, Assistant Professor, Instrumentation and Control , Ld college of engineering, We express our gratitude towards our parents for their continuous support, understanding and patience without whose support and understanding this endeavor would never have been fruitful. We also thank all our friends for helping us out during the downs. We totally respect everyone for giving their time when we were in need.

Thank You.

ABSTRACT

ATCS usually known as "**Automated Toll Collection System"** is a new technology for collecting toll in a faster and more efficient way. It is a great alternative to long waiting at manual toll plazas. In order to overcome the wastage of time and fuel at same time we have come up with a concept of RFID based automated toll collection system using Arduino. RFID stands for **Radio Frequency Identification**; these cards are unique identities provided to every vehicle by Registration Office at each city. Whenever a vehicle with such Unique ID reaches the toll plaza, the RFID card reader attached on the toll plaza gate reads the card and transfers the unique ID to **Arduino ATmega 328.** Accordingly, the processor works and deducts a fixed money from the prepaid card. If the card's ID is valid and has sufficient balance, the processor will command the **DC motor** to start and open the gate, letting the vehicle to pass. On passing from the gate the screen will show the deducted and current balance of the card. If the card is not valid or low in balance, it will prompt to move the vehicle to manual toll collection lane.

Furthermore, a message will be sent to the owner's registered mobile number with the same details along with the location of the toll booth.

Keywords: ATCS, RFID, ATMega328, Arduino, SIM900.

Index

Chapter No.	Title	Page No.
	Acknowledgement	**ii**
	Abstract	**Iii**
	Index	**iv**
	List of Figures	**V**
	List of Tables	**vi**
	List of Abbreviations	**vii**
1	**Introduction**	
	1.1 Toll Collection System	1
	1.2 Electronic Toll Collection System(ETC)	2
	1.3 Embedded Systems	3
	1.4 Microcontrollers	3
	1.5 Radio Frequency Identification(RFID)	4
	1.6 Project Preview	5
2	**Literature Review**	
	2.1 Literature Survey	6
	2.2 History of Toll Collection	8
	2.3 Problem Definition	10
3	**Materials and Methods**	
	3.1 Hardware and Software Requirements	16
	3.2 Methodology	25
4	**Results and Discussion**	
	Example-1 RFID Reading Output	39
	Example-2 SMS sent using GSM Module	40
	Example-3 Money Deduction Output	41
	Conclusion	45
	Summary	46
	References	48

LIST OF FIGURES

Figure.1.1	A Manual Toll Collection System in India.
Figure.1.2	An Electronic Toll Collection System
Figure.1.3	The RFID Toll System
Figure.1.4	Arduino and EM-18 Interfacing
Figure.2.1	Passive RFID Tag
Figure.2.2	Atmega328 Arduino Board
Figure.2.3	Toll Collection Booths
Figure.3.1	Circuit Diagram Showing all Hardware Used
Figure.3.2	ARDUINO Board with ATmega 328
Figure.3.3	EM-18 RFID Reader
Figure.3.4	RFID Tags
Figure.3.5	Motor Shield
Figure.3.6	9v DC Battery
Figure.3.7	LEDs
Figure.3.8	SIM900 GSM Module
Figure.3.9	Jumper Wires
Figure.3.10	The “V” Diagram
Figure.3.11	RFID Working
Figure.3.12	Flow Chart of Project’s Procedure
Figure.3.13	RFID Reader Interfaced with ARDUINO
Figure.3.14	ARDUINO, GSM Interfacing
Figure.3.15	ARDUINO, EM-18, Basic Shield & Motor Shield Interfaced
Figure.3.16	ARDUINO, EM-18, Basic Shield, Motor Shield & GSM Module
Figure.4.1	RFID Reading Output
Figure.4.2	GSM Module SMS Sent
Figure.4.3	Money Deduction Output
Figure.4.4	Received Test Message
Figure.4.5	Money Deduction Message

LIST OF TABLES

Table 1	Vehicles Passing a Toll Booth Year-Wise
Table 2	Total Fuel Consumption
Table 3	Hardware Description
Table 4	Comparison of Different ATCS Techniques

LIST OF ABBREVIATIONS

ATCS:	Automate Toll Collection System
ETC:	Electronic Toll Collection
RFID:	Radio Frequency Identification
GSM:	Global System for Mobile
CPU:	Central Processing Unit
LED:	Light Emitting Diode
LCD:	Liquid Crystal Display
RAM:	Random Access Memory
ROM:	Read only Memory
EEPROM:	Electronically Erasable Permanent Read Only Memory
DC:	Direct Current
SIM:	Subscriber Identity Module
TCP/IP:	Transmission Control Protocol/ Internet Protocol
GPRS:	General Packet Radio Service
GNSS:	Global Navigation Satellite System

Chapter 1

INTRODUCTION

1.1 Toll Collection System

While driving along Highway, we all encounter single or multiple booths in which we have to pay a fixed amount of money. These booths are known as toll booths and the money we pay is the tax for using the road known as the toll road or toll way. Since, most of the roads are built with the money that is raised by state or national government through the taxes. So, toll is a kind of tax that we pay to the government for the maintenance of the highways.

1.1.1 Methods of toll collection

There are basically two methods of toll collection:

1. **Manual Toll Collection:** This method is the most widely used method in India. In this, when the vehicle arrives at the toll booth the driver gives the money to the collector at the booth who feeds the information in the computer and provides a receipt to the person in the vehicle. Each vehicle has its amount fixed.

   For eg: a two-wheeler driver needs to pay Rs 20, for car it is Rs 40 and for larger vehicles like buses and trucks its more. But these amounts vary from state to state.

2. **Electronic Toll Collection:** As the name suggests, the whole system is automatic and no human efforts are required. It is not a much familiar method in India but is commonly used in countries like USA. It uses a card or a tag to identify the vehicle and automatically deducts the amount from the bank account. It is much faster way.

Fig.1.1 A Manual Toll Collection System in India.

1.2 Electronic Toll Collection System (ETC)

In the era of modernization where everything is being automated or made electronic, toll systems have also undergone great changes. From past few years, the electronic toll systems have come under use and are becoming famous because of the benefits they have over the manual toll systems. The ETC systems, unlike a manual system do not involve a collector. The vehicle when arrives at the booth it does not need to stop at the booth like it had to do in manual system. It directly passes away without giving cash to anyone. Thus, we can call it a cashless toll collection.

The ETC system involves various components which work together and form the whole system. As the vehicle arrives at the booth a sensor which has been incorporated at the booth, senses the tag or a card embedded in the vehicle. This tag is known as a RFID card. This RFID card has a unique identity for every user and thus has the information regarding the user. The system reads the card and authenticates the person to pass through the toll after deducting a fixed amount from the user's bank account. The payment in the ETC system is made through wireless mode. There is an antenna at the tollgate which establishes a wireless connection with the on-board device when the vehicle's RFID card is sensed and thus, automatically deducts the amount from the account. There is no need for the vehicles to stop and wait for the process to happen. It occurs in few seconds and thus, it is a very effective and a fast system.

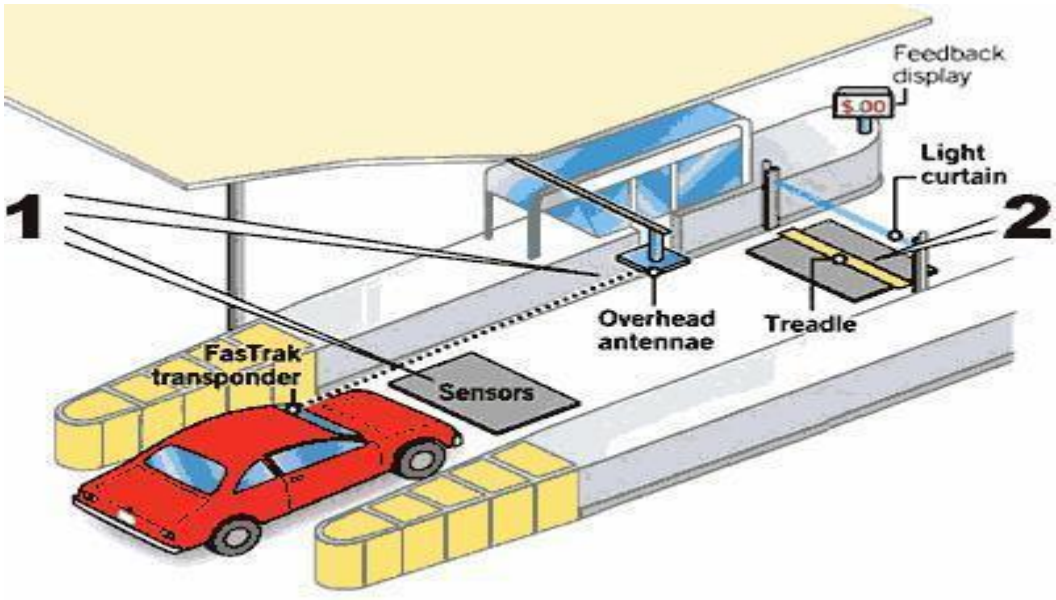

Fig. 1.2 An Electronic Toll Collection System

1.3 Embedded Systems

The Embedded Systems are the computer systems that are designed for any particular function or task. These systems are the combination of hardware as well as the software. There are various systems around us which have electronic components embedded in it. Almost all the electronic devices work on the principle of Embedded Systems. Whether it is a washing machine, a camera, various medical and industrial equipment or a mobile phone, all of them have some electronic device embedded in it. Some of the systems are programmable and some are not. Programming a system provides flexibility of using the device for different purposes. Embedded Systems find a very large variety of applications due to the increased use of electronic devices and machines. Similarly, one application of embedded system is in Electronic Toll collection systems. The ETC systems have surveillance cameras, sensors, displays, computers at the control units and all of these are controlled by the embedded systems. They all function due to embedded systems.

The moment the RFID tag is sensed by the sensor the processing of the information stored in card begins. The information is matched with the information stored as a database in the computer. If there is a need to give any penalty, it is also determined by the system, and then the gates get open after the payment. This whole process of sensing, checking, payment, toll gate operations are performed by the Embedded Systems.

1.4 Microcontrollers

A microcontroller is a computer designed to perform particular tasks which appear to be small or minor but are as important in a system. For Eg: turning ON an LED. A microcontroller comprises of the processor or the CPU which performs the task for which it is designed. It has the input and the output ports to which any external device can be connected. It also has its memory i.e. RAM and ROM which means that data can be stored in it or processed value can be stored. One of the major use of microcontrollers is in embedded applications. Thus, it is very useful in ETC systems.

1. Reading information and the RFID card number from the RFID card of the person.
2. Sending this data on the LCD screen of the operating person.
3. Sensing and receiving the signals from the IR receiver.
4. Sending the signal to the motor to open the gate if the data is genuine and the card has the balance or else sending the signal to the buzzer if any of case does not apply.

1.5 Radio Frequency Identification(RFID)

It is a new identification technology that uses radio frequencies for identifying any object or a person. It's one major advantage is that it is wireless and no one can read the information stored in it accept for the device meant for reading it. Thus, the information is confidential. It is generally done using a RFID tag which is a small card with an electromagnetic chip embedded over it with an antenna. All the information is stored in that chip. Generally, a RFID serial card number is used as the identification number. Each user has its unique ID number stored in that chip. A RFID system comprises of the RFID tag and the reader or the sensor. Both the card and the reader have their antenna for sending and receiving the signal. There is a specific range in which the tag works with the reader. As soon as the tag reaches the range of the reader, it gets induced and sends the information.

1.5.1 Working of RFID

When the vehicle arrives at the toll plaza, the RFID tag attached to it is sensed by the sensor. The moment the tag comes within the magnetic field of the RFID reader, the current is induced in it and it gets energized. Then, it sends the information stored in the chip in form of the radio frequency signals through the antenna. When the antenna of the reader receives the signal, it again converts it into digital form thus making it suitable for displaying on the LCD screen of the operating person.

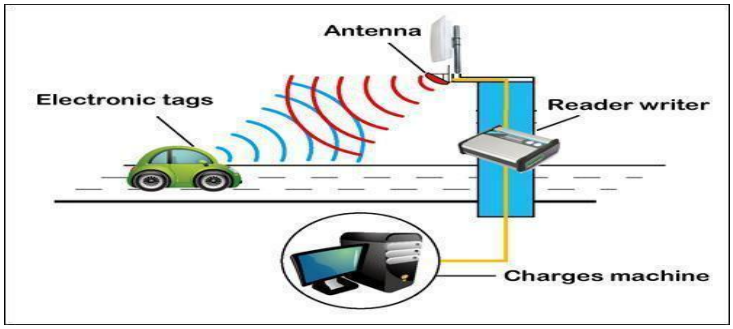

Fig. 1.3The RFID Toll System

1.6 Project Preview

This project is based on the Electronic Toll collection systems. It has been programmed using Arduino. Arduino is open source software which is easily available for use and its modifications. It consists of a programmable circuit board and software, IDE (Integrated Development Environment). This software is used to write a code and then feed this code to the physical circuit board for a desired function or a task. It is very user friendly and

easy to use. Unlike other software, it does not require any additional circuit or device to load the code in the circuit board. It just needs a USB cable. The software can be run on our computer systems and from there the code can be transferred easily.

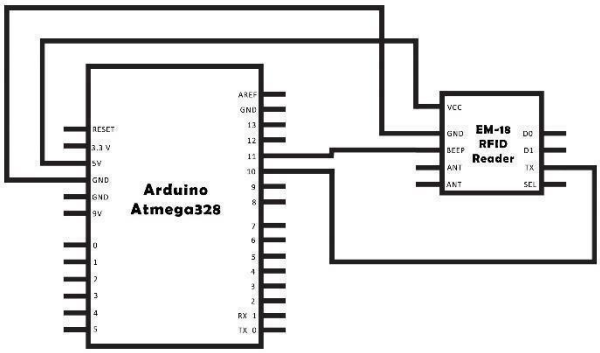

Fig. 1.4 Arduino and EM-18 Interfacing

This project works like this:

1. A car will approach the toll booth. The RFID reader at the toll will read the RFID card embedded on the vehicle.
2. Then, it will pass the information stored in the card to the Arduino.
3. The Arduino will track information and will check whether the card is valid or not.
4. If the card is valid, the owner's name will be displayed on the screen. A certain amount will be deducted from the owner's account. Here, we have fixed the amount for different users. For Jenil, the amount is Rs 50 and for Yash it is Rs100. Along with this, a green LED light will glow twice with the buzzer doing the same.
5. After this, a command will be send to the motor to open the gate and the vehicle will pass. Now, GSM module will send message to the owner informing him about amount deducted from their tag. Here, message will be: 'Rs 50/100 deducted'
6. If the card is not valid, the screen will show 'invalid user' and no amount would be deducted. Along with this, a red LED light will glow with the buzzer making noise continuously for 2 seconds. Thus, no command will be send to the motor.

Chapter 2

REVIEW OF LITERATURE AND DEFINITION OF PROBLEM

2.1 Literature Survey

The Toll Collection System has changed drastically over the years; from being a single borderline, a small passage booth to the huge toll collection infrastructure that has a key role in the revenue generation as well as the working of the traffic of a city or even a state. While the majority of the population is travelling via different means it has become a necessity and a regulatory method to control traffic.

The following papers were referred and reviewed in order to understand the role of Toll Collection System:

Kamarulazizi & Ismail, 2005 [1] the paper mentioned the benefits of Electronic Toll Collection System often abbreviated as ETC over the Manual Toll Collection System. It stated that the former helps in controlling the congestion that is generally caused by the traffic during the festivals. Moreover, it is also beneficiary for the operators as it helps in Controlling the Audit which is centralized.

Chhoriya, Paliwal & Badhan, 2013 [2] it was presented in the paper that the use of image processing tools can be used for the toll collection. The number plate on the vehicle can be processed and checked so as to match any existence in the database; from which the toll can be deducted thus giving it an upper hand from the manual toll collection.

Salunke, Malle, Datir & Dukale, 2013 [3] the following paper conceptualised the use of RFID i.e. Radio Frequency Identification for the toll collection thus making the whole system easy and feasible. The use of ATCS expanded to Automated Toll Collection System is an effective way that could lead to reducing the losses the Government faces. The idea proposed in the paper had an edge over other electronic methods due to its flexible nature and easy implementation.

Kalantri, Parekar, Mohite & Kankapurkar, 2014 [4] with a slight change in the existing system, the paper proposed the use of passive RFID tags that can be used instead of active RFID tags that takes use of the car battery. Furthermore, it also explained the usages of the proposed system Vehicle Theft Detection, Signal Breaking Avoidance and Tracking of Over-Speeing Vehicle.

Aphale, Chaudhari & Bansod, 2014 [5] the paper introduced the use of a microcontroller ATMega328PU which is the centralized unit that controls the proposed system that uses RFID to detect any close range tags and hence requesting the database to deduct amount from the account connected to the ID.

2.1.1 Motivation of Project

The certainty of any system is based on the hardware and the selection of the same is a great deal. But it also takes the lead once the ideas have been generated; after reviewing the base papers the following questions were derived:

1. What should be technology that can be used for Electronic Toll Collection System?
2. Is there a scope for any improvement if the following work has been done?
3. How beneficial can the work be?

Fig 2.1 Passive RFID Tag

The answers to these questions were a fix that provided the project a platform to build on. Firstly, the use of RFID or Radio Frequency Identification was decided due to its easy implementation and flexibility to adapt to any existing system or technology. RFID tags are placed or attached to a vehicle that can be traced and read by the RFID Reader. The reader mentioned is an active device that needs a power supply of its own whereas the tag is a passive one which contains a unique ID that can be read. The technology comes in different ranges of frequency thus falling under main Radio Frequency Band.

Furthermore, the use of passive RFID tags is a better option than going for an active one as the implementation is easier and won't require any prior usage of power from the vehicle's battery. This system helps to fasten the toll collection as it reduces the waiting time on each toll collected and at the same time [6] is hassle free and more user friendly than any other proposed system till date.

Aphale et al., 2014 stated the use of a microcontroller named ATMega328 which can be simplified by the use of the same controller mounted or embedded as an Arduino Board. This Arduino board will be regarded as the microcontroller and hence work as the central hub for the processes taking place in the system. It would contain different pins to accommodate the various components required, take power from the user defined source and control the output as desired.

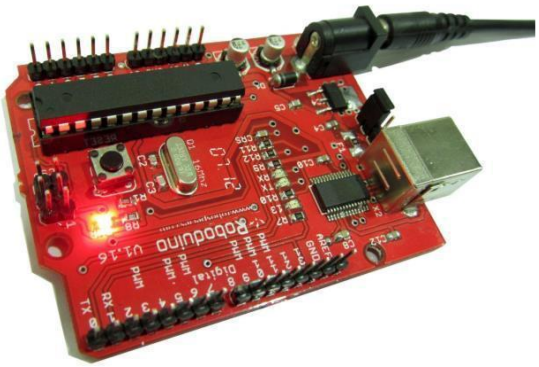

Fig 2.2 Atmega328 Arduino Board

2.2 History of Toll Collection

The following secondary study or research was done to understand the basic requirements and key study of the previous and present systems in existence. The study involves the basic queries like what is toll collection, why is it done, current toll collection systems, problems with each toll collection system and in the end the final proposed system for this project.

2.2.1 Toll Collection

Toll, turnpike, tollway or toll fee are the generic terms for a certain amount or fee that is taken to assess the passage. This fee is taken at stops or points which are called Toll Booth, Toll Plazas etc. This booth has a small shed that is manned and the individual responsible collects the cash, change from the user who travels via that passage and does the data entry for the vehicle followed by opening the gate for the vehicle to pass. This method is called the manual method of toll collection. Some other methods of toll collection include the unmanned toll booths consisting of a device/ machine that takes in coins and money upon which it opens the gate through which the person can enter with the vehicle. After some up gradation the automated toll collection systems have been developed which contains new technology based devices that open the gate and let the user pass only after identifying the vehicle and its license plate and registration number.

Fig 2.3 Toll Collection Booths

2.2.2 Need for Toll Collection

This fee or toll is taken in order to meet the cost of construction of roads, signs and other maintenance activities on the public roadways. This toll fee is often accounted as a revenue that is paid by the general public.

In India, when an individual pays a toll tax it is forwarded to the State Government which is totally different from the taxes that are paid monthly or annually. This is due to the reason that the infrastructure used in the roadways and other related activities is a joint venture between a private organization and the Government. When the contract is

signed it is mentioned that the toll will be collected up to a certain amount that could meet the goal set by the private company or organization. After that no toll or fee is taken. This is also done for the same reason as mentioned above.

One other reason of choosing toll collection is to control the demand and congestion management. The more the vehicles pass through the toll booth, the more the toll is increased so as to control the traffic entering a certain place; along with this certain toll booth are also placed to regulate the use of public transport.

2.3 Problem Definition

2.3.1 (a) Feasibility Study

In [3] the study of the feasibility reports the very need of improvement in the toll collection system. It states that if there are in total 100 manned/manual toll booths and if every day about 100 vehicles pass through each system then; the count of total vehicles passing one such tool booth per year would be about 36,000 (100 x 30 x 12).

Similarly, for 100 systems, the number goes high up to 36, 00, 000. For simplification, the table shows the same data as mentioned above:

Table 1. Vehicles Passing a Toll Booth Year-Wise

Number of toll booths	Days	Number of vehicles
1	1	100
1	30 x 12	36,000
100	30 x 12	36,00,000

This table shows that per year nearly 36,00,000 vehicles remain in the still position for nearly 6 hours in a condition where their engines are roaring and consuming fuel, at the same time creating air pollution. Now if we suppose that in that 6 hours a certain vehicle uses about 1 L fuel; the total volume of fuel for all those 36,00,000 vehicles go up to the same amount i.e. 36,00,000 L.

Table 2. Total Fuel Consumption

Total Vehicles	Amount of Fuel Consumed	Total Amount
1	1	61 Rs.
36,00,000	36,00,000 L	21,96,00,000 Rs.

The current price of fuel is nearly 61 Rs. So assuming that 36,00,000 vehicles pass through all the toll booths nearly 21,96,00,000 Rs valued fuel is wasted during that stoppages at the manual toll booths.

This data is staggering considering the fact that all the data taken are minimum and the result is out of expectation.

2.3.1(b) Need for upgradation

There are many reasons behind the need for an upgradation in the manual toll collection system. Some of them are:

1. Manual toll collection system is slow as it ceases the vehicle flow as each vehicle approaches the toll booth for cash transfer and entry into the database. This leads to various issues like fuel wastage, time wastage and air pollution.

2. Manual toll collection system is often found guilty of embezzlement as the toll booth operators who are in charge of the same are involved in the theft of toll books and cash. This and other cases lead to this type of collection system not secure in terms of money.

3. The manned systems often see themselves vulnerable to harassment by the vehicle owners. Mostly in India, where such cases have been found that the toll operators are physically harassed and beaten for they have done nothing wrong. The use of ETC can curb this problem single handed.

There are other systems that also need upgradation due to one or other fact that hinders their working:

1. The Bar Code System used earlier was based on reading a bar code that acts as a Unique ID for the user's vehicle, but the issue with the same is that the bar code reader is not that accurate when it comes to reading the code. There have been many cases where the readers have not operated properly thus making the whole system manual again.

2. The Image Processing System; which dealt with the number plate recognition also faced similar problems, but instead of the readers going shut, they have been found inefficient in reading the number plates and hence giving false alarms of the vehicle not being registered.

RFID Active Tag System which as mentioned above uses the car battery power to work is a total low for a feasible system which can be easily neglected by the users and replaced by the manual toll collection system.

All these problems have pointed towards a solution that requires the system to be a fully functional, user friendly system that not only meets the demands of both the users, but also of the toll collectors. The system is supposed to be:

1. Feasible
2. Easily Implemented
3. User Friendly
4. Smart System

2.3.2 Proposed System

The system proposed in this paper meets the above requirements and hence can be considered a solution to all the problems that the previous systems have undergone. The proposed system uses the same RFID or Radio Frequency Identification Technique for identifying the vehicle and checking it against the database. The same has been done but the use of passive IDs can be considered better instead of using the non-feasible active devices or IDs. These passive devices will be used in the form of stickers or thin cards that would be either placed on the windshield or the side portion of the vehicle; that totally depends on the RFID Reader placement. These passive devices are made up of coiled antennas that are connected to a small data/ memory device, when an active device (here: the reader) approaches the passive device it sends radio waves that when contacts the passive device create a magnetic field. The same draws out power from these waves and sends the encoded data in the memory to the reader, which in turn sends it to the main system or the database.

In the proposed system the use of embedded system has been done thus the application of Arduino ATMega328 is prominent. When a vehicle with a passive RFID Tag approaches the toll booth, the RFID Tag Reader reads the data from the tag and sends the Unique ID to the main system i.e. the Arduino device that checks it from its database,

and thus responds to it by displaying whether it is a registered vehicle or the opposite. If it is former, the Arduino alarms the led and the buzzer to turn green twice and beep simultaneously. The system then asks the toll gate to open by powering up the motor connected to it for a certain period of time. As the vehicle passes away the gate is closed.

As all this happens, a certain toll fee also gets deducted from the user's account in a certain bank that is connected to the database of the system. There are two such conditions that fall under this process: The first being ample amount to deduct the fee, if yes, the vehicle is allowed to pass whereas not meeting the same leads to the second condition where the cash being less in the account; the vehicle is asked to move for the manual toll collection.

If at first the reader passes the ID and the Arduino signals that the vehicle is unregistered, the vehicle is asked to move to the manual toll collection.

Apart from all this, when a registered user/vehicle passes a toll booth/ toll plaza the GSM Module connected to the toll plaza sends a message to the user's registered mobile number giving details about the amount deducted from the account for the registered vehicle number along with the toll booth that the vehicle just passed.

Advantages of the proposed system:

1 The proposed system is easy to implement and can be placed and easily mounted onto the toll booth's main setup.

2 The built cost is very less as compared to any other toll collection system.

3 It is faster, when it comes to reading tags, moreover it being a hassle free system has no or less drawbacks.

CHAPTER 3

MATERIALS AND METHODS

This project was proposed considering the fact that, these days' transportation is a major part of human day to day life. Since, time and money are two major components of transportation so, even if is possible to save a small amount of both entities at same time it would be a boon to the society and the environment. The idea of ATCS (Automated Toll Collection system) is already being used in a lot of countries worldwide. India being a country with 2nd largest population in the world and a country with 2nd largest chain of highways must have ATCS implemented all over India to resolve the problem of jams at toll booths. This project is a working model of the highly efficient and cheap system which can be used on toll plazas. ATCS uses RFID technology for identification of vehicles passing through toll plazas and an ATmega328 microcontroller embedded in Arduino board for further processing.

This automated RFID system can also be use at places like automated parking, security control, vehicle tracking etc. The basic methodology for this project is designed so that this system can be available to everyone who needs it and can be installed anywhere by anyone. The principle of this project is RFID i.e. Radio Frequency Identification technique used to find which vehicle has reached the toll and using its pre fed information to carry out further functions like allowing the vehicle to pass or not. ATCS is mostly dependent of its microcontroller i.e. ATmega328 embedded on ARDUINO board. The owner of the vehicle has a RFID tag, which is read by the low frequency RFID reader to read even tags of 125 kHz frequency. On reading the tag, the tag ID obtained is sent to microcontroller after which it analyses the tag ID and matches it with the IDs saved in database, if the vehicle's tag ID matches with one of them it deducts a certain fixed amount of money from the prepaid card and gives a command to DC motor to open the gate and allow the vehicle to pass.

If in case the card is not valid the microcontroller commands the owner to move the vehicle to the manual tax payment lane. In case the card is valid but the money in the card is insufficient same command is given to the owner. On successful processing or transaction of money the owner receives an automated SMS through GSM module

SIM900 on registered mobile number regarding the amount of money deducted, the available balance and the name of the toll plaza vehicle recently crossed. This feature helps the owner to keep a track on his vehicle all the time preventing the theft of vehicles. Basically, ATCS discussed in this project contains some of these hardware components shown in this circuit diagram:

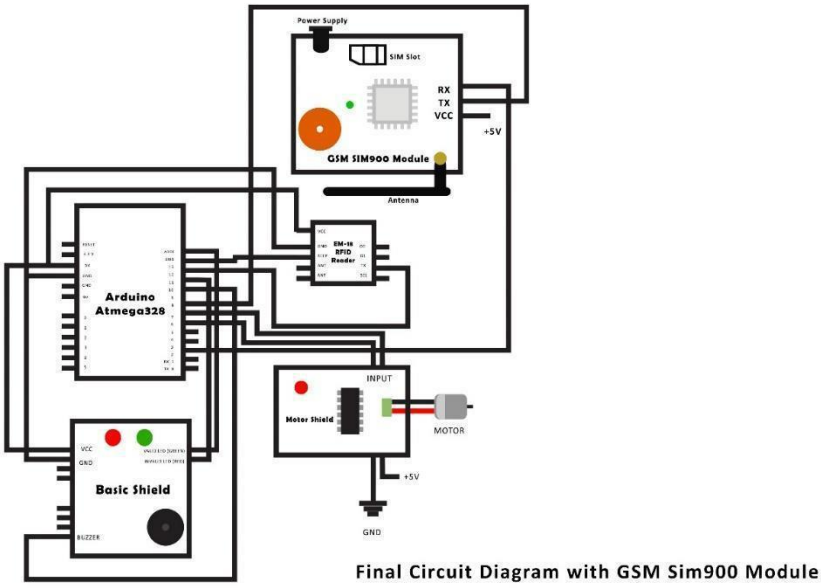

Fig. 3.1 Circuit diagram Showing all Hardware Used

1. ARDUINO Board with ATmega 328 microcontroller
2. EM-18 RFID Reader
3. RFID cards (125 kHz)
4. LEDs
5. Buzzer
6. DC Stepper Motor
7. 5v DC Power Supply
8. SIM900 GSM Module
9. Jumper Wires
10. Bread Board

There are certainly a lot of methodologies or ways to design a ATCS which include Active RFID, Passive RFID, GNSS, Infrared technology, Microwave technology, but this projects demonstrate the best one of all these i.e. Passive RFID technology. Passive RFID means Radio Frequency Identification device which do not require power source to activate, it works automatically and is detected without the help of any power source.

3.1 MATERIALS

3.1.1 HARDWARE REQUIREMENTS

- ARDUINO Board with ATmega 328 microcontroller

The ARDUINO is an electronic multipurpose board based on ATmega 328 microcontroller. It has 6 Analog input pins and 14 digital IO pins, out of these 14 pins 6 can be used as PMW outputs as well. It contains a lot of components along with the microcontroller like16 MHz quartz crystal, a reset button, a USB connection, an ICSP header, a power jack and lot of electronic components. It already has all the things embedded in it which are needed for proper working of the microcontroller. It can be directly connected to the computer through USB cable or can be used either by giving power through AC/DC adaptor or through a battery.

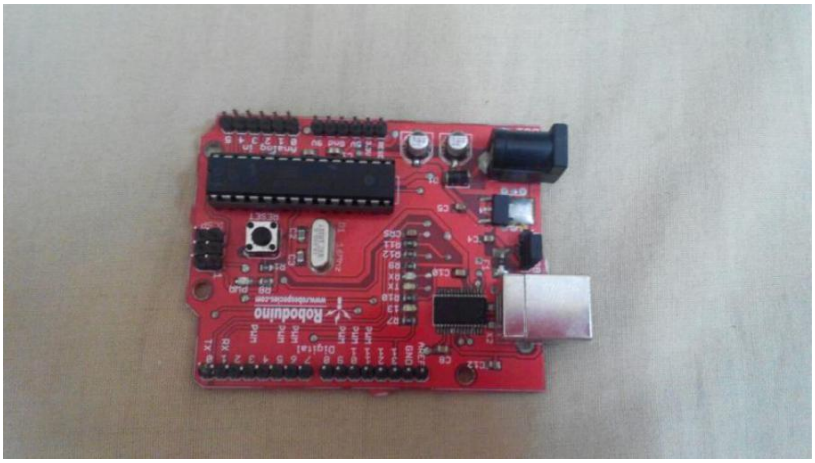

Fig. 3.2 ARDUINO Board with ATmega 328

Key Specifications are as follows:

1. Microcontroller: ATmega328
2. Operating Voltage: 5 volt
3. DC current per I/O pin: 20 Ma
4. Flash Memory: 32 KB
5. EEPROM: 1 KB
6. Clock Speed: 16 MHz

The ATmega328 is a unity chip microcontroller manufactured by “Atmel” in its megaAVR chip family. This microcontroller is usually used in projects and automated systems in which cheap, simple and less powered microcontrollers are needed. On the other hand, it is widely used in ARDUINO boards like ARDUINO UNO and ARDUINO Nano etc.

- EM- 18 RFID Reader

RFID reader is an active device with is powered through the Arduino board. It has an inbuilt small antenna in it, which emits radio waves continuously when it is in active mode, and the RFID tag responds to the radio waves by sending its data to the reader. In this project the use of RFID reader is to detect the tag on the vehicle arriving at the toll plaza and providing the tag’s ID to the microcontroller.

EM-18 RFID reader has some specifications such as:

1. Operating Distance- 10 cm
2. Operating Voltage- 5 volts
3. Operating Frequency- 125kHz
4. Current Consumption- <50 Ma

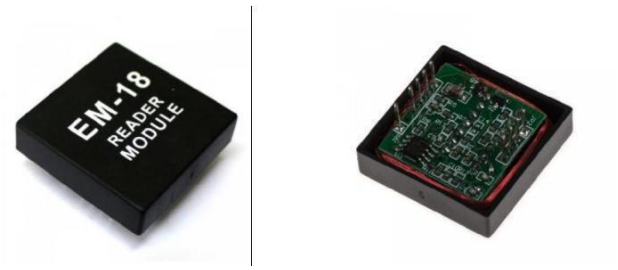

Fig 3.3 EM-18 RFID Reader

In this project it is used as stationary reader which is always looking for the RFID tags in its range. If any vehicle with RFID tag comes in its range it receives its data and transfers its unique ID to the ATmega 328 for further functioning.

EM-18 RFID reader is basically microcontroller-based transceiver; it gives power to the tag with help of EM radio frequency or known as RF field. As soon as the RF field passes RFID tag’s antenna, alternating current voltage generates and this AC voltage is rectified to input power of RFID tag. On getting powered from reader, the tag is able to get commands from EM-18 RFID receiver. The hardcoded data of tag now can be read by RFID reader and sent to the microcontroller for processing.

- RFID Tags

A Radio Frequency Identification tag/card usually known as RFID tags are electronic gadgets that can be attached with a product, person, anima or many more for their identification or tracking using radio waves. Every tag its own identity or tag number, these tags are of different types, some can be read from long distance and some can

even be read from a distance place up to which a normal human cannot see. Most of these RFID tags are made of two parts, one is an antenna for transmitting and receiving of signals, and other is integrated circuitry for processing and storage of information, it can also be used for specific functions.

Fig 3.4 RFID Tags

There is a technology called chip less RFID, which helps to identify the tags without the integrated circuit in it, this making of tags without integrated circuitry helps in lowering the cost of RFID tags than traditional ones. These days there is a significant uplift in RFID usage, helping in improving the efficiency of product tracking and managing things. In other words, RFID tags are automated identification object or transponders.

There are basically three types of RFID tags:

1. Passive tags, with no power supply,
2. Active tags, with power supply, and
3. Beacon tags.

The RFID tags use in this project are passive type this helps in saving power and has no problem of providing power supply to the tags all the time to make them work. Passive tags work with the help of small amount of current induced in tag's antenna from RF signal of reader, this current generated enough power for circuitry in tag to start up and give response to the reader.

- DC Stepper Motor

The motor use in this project is a battery operated DC motor usually called stepping motor or step motor. They are so called because of their functioning, i.e. since these type of motor rotates a fixed angular step for every i/p current pulse got from controller. The market of these motors is on hike due to growth in robotics industry. They are better than others as they can be easily operated directly through computers, microcontrollers and in our case through ATmega 328.

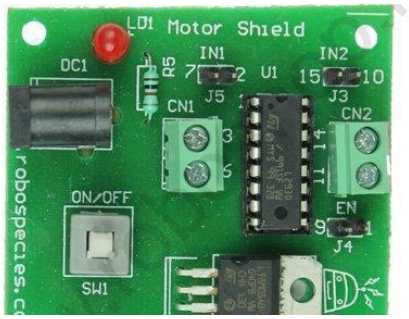

Fig 3.5 Motor Shield

In this project, work of these stepper motors is to lift toll gate to let the vehicle pass on receiving a positive signal from the microcontroller which is generated when RFID tag is valid and the transaction is successful. Stepper motor rotates 90º degrees anticlockwise to lift the gate then waits in that position until the vehicle passes and then rotate 90º clockwise to close the gate.

- Power Supply

The power supply used is 5 volts to 9 volts DC power supply through DC batteries. The current used is 2 mille Ampere. Since the whole project is power dependent it is important to keep fixed power supply to do that we have in build regulators in Arduino board and all other shield's like MOTOR shield and BASIC shield.

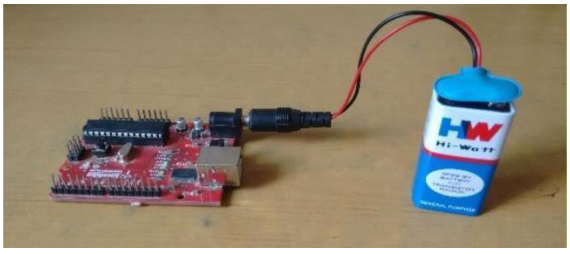

Fig 3.6 9v DC Battery

The ARDUINO board has integrated capacitor to store power and release regulated supply as needed by different components. This project uses two 5 Volt DC batteries one to power up the ARDUINO board and other for the motor to open toll gate.

- LEDs

Light Emitting Diode or IR LEDs is also one of the important component used in this project, basically it is used as visual indicator to the owner of the vehicle weather his/her RFID tag is valid to let him pass the toll plaza or not. There are two LEDs used in this project a green colored LED and a red colored one. The working principle of LED is electroluminescence, they have two semi-conductor light sources which when activated or given electricity produce light and the LED glow.

Fig 3.7 LEDs

This project uses a Basic shield which had a green and a red LED inbuilt, when a vehicle with RFID tag reaches the toll booth the RFID reader reads the tags ID and passed information to microcontroller for further processing which carries out following functions:

RFID Tag: If Valid: Green LED Blinks twice
If Invalid: Red LED Blinks twice

- Buzzer

The buzzer is an electronic device which creates a beeping sound when turned on or activated. A buzzer can be mechanical, piezoelectric or electromechanical. This project uses an electromechanical buzzer to indicate the state of the card. This buzzer can be turned on or off by just giving an electrical pulse from the microcontroller.

In this project the buzzer is used as an indicating device in following ways:

RFID Tag: If Valid: Buzzer beeps twice with a short interval of time, and

If Invalid Buzzer beeps continuously for 5 Seconds

- SIM900 GSM Module

This is one of the most important electronic module used in this project, to let know the owner of the vehicle about the transaction taking place through his vehicle's RFID tag with the help of a Short Message Service i.e. SMS. It uses GSM technology also known a Global System for Mobile Communication to send the SMS to the registered mobile number of that RFID tag.

Specifications of RFID Module:

1. Quad-Band GSM with 1900 MHz frequency
2. Built in RS232 Level Converter
3. Configurable Baud Rate
4. SMA connector with a GSM L Type Antenna
5. SIM card holder
6. Built in Network status LED
7. Strong TCP/IP protocol stack for internet data transfer using GPRS
8. Operable under -20º to +55º C
9. Voltage: 5V – 12V DC supply

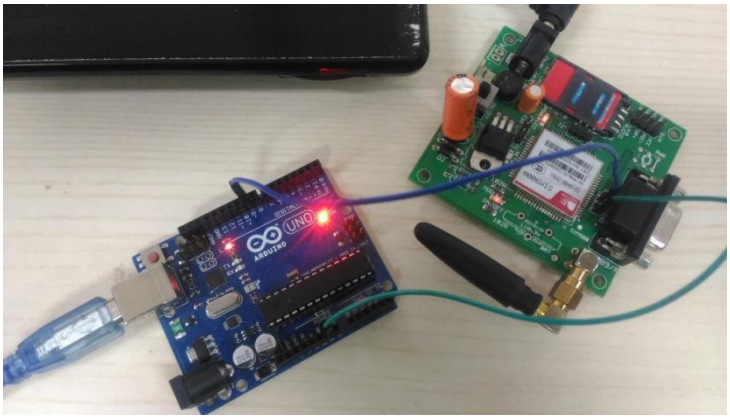

Fig 3.8 SIM900 GSM Module

The SIM900 GSM module used in this project is one of the most use module for this kind of works. In this projects its function comes after the vehicle has crossed the toll

booth, on successful transaction of the money from the owner's card, it sends an acknowledgement in form of an automated SMS to the registered mobile number to that card, informing the owner about the money deducted current balance and the information of the toll plaza where transaction was done. This added feature helps he owner of the vehicle to have a track on its vehicle in case he has handed it over to someone.

- Jumper Wires

As always jumper wires have been used in this project to interconnect the different modules to each other like connecting Arduino to DC motor, connecting Arduino to GSM module interconnecting Arduino, GSM module, motor shield and basic shield with the help of bread board. Basically, a jump wire is a small electrical wire having a solid tip at both ends to connect different electronic components to each other.

There are basically 2 types of jump wires:

1. With Crocodile clips
2. With Insulated Terminals

There are further sub-divided into 3 categories:

1. Male to Male
2. Male to Female
3. Female to Female

Fig 3.9 Jumper Wires

The jumper wires used in this project are male to male, male to female, female to female, all insulated terminals one.

- Bread board

Breadboard is a construction platform for prototyping of circuits and systems. The breadboard used in this project is a solderless breadboard to make the entire circuit of project at initial stages.

Summing up the hardware used in this product, below shown is a table of hardware used with their specifications:

Table 3. Hardware Description

S.no.	Hardware	Cost(Rs.)	Model	Specification	Function
1.	Microcontroller	350	ATmega328	28 pins, 5v, 16MHz	Controlling all the Connected modules.
2.	RFID Reader	450	EM-18	9 pins, 5volt	To read the nearby RFID tag.
3.	RFID Tags	30/-	Card type	125KHz, 4 inch dist.	To provide unique ID to every vehicle.
4.	GSM module	1200	SIM900	1900 MHz, 9 volts, SIM slot.	Send SMS to owner About transaction.
5.	LEDs	10/-	_	3volts, 2mA	Indicator
6.	Buzzer	60	Electro-mechanical	5volts, Beeping	Indicator
7.	Battery	30/-	Alkaline	9vols, DC	Power Supply
8.	Motor	200	Stepper	5volts, DC	To control gate
9.	Jump wires	5/-	Insulated	_	Connectors
10.	Bread Board	160	_	_	To design circuit

3.1.2 SOFTWARE REQUIREMENTS

Arduino Compiler – Version: 1.6.8 for Windows
Source: www.arduino.cc

Cost: Free

I t is an open source software available directly at Arduino's website, for beginners and professionals to carry out projects like this as well as more projects and stuff.

Language Used: Embedded C

Embedded C is a very similar language to C and C++ with few of its libraries and function name being different it serves as an easy and simple language to make embedded systems work according to the coders need.

This was all about the hardware and software used in this project, all electronic equipment, components and software used are of specific parameters and used for specific functions. Each device has its own importance and value in this project.

Concluding the chapter of this project, there are some facts regarding the overall software and hardware used in this project which are as follows:

Hardware used: 10
Source: Online Shops
Average Cost of items: Rs.270
Overall cost: Rs. 2700
Software Used: 1
Source: www.arduino.cc
Average cost: Rs. 0
Overall cost of software: Rs. 0 (Free)

The final cost of this project i.e. ATCS (Automated toll collection system) is around Rs. 3000, which makes it easy and affordable for everyone to install it anywhere they want.

3.2 METHODOLOGY

The basic methodology used in this project is integration of small Embedded C codes for each hardware to form a large working code for a working model of ATCS (Automated Toll Collection System). Embedded C is a very similar language to C and C++ with few of its libraries and function name being different it serves as an easy and simple language to make embedded systems work according to the coders need.

The basic methodology or concept that is developed to make this project is explained through this "V" diagram:

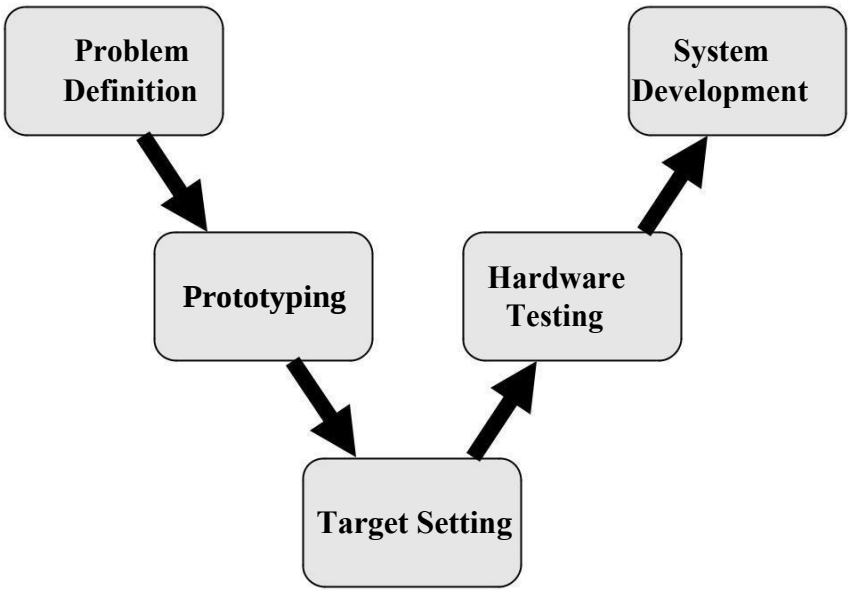

Fig 3.10 The "V" Diagram

As shown in V diagram first step taken was defining the problem, or we can say finding out what's the problem in manual toll system, which has already explained was lot time consuming and fuel wasting method, so to solve these problems ATCS is introduced. Secondly, prototyping the project or the circuitry, which is done through basic hardware available. Thirdly, target setting, or goal, then the goal for final look or working of project was set.

Fourthly, setting up hardware at initial stages and then interfacing hardware with each other for testing at later stages. Finally, developing a proper system for the hardware and the code to make the project look like a proper working model of ATCS.

The basic methodology governing this project is the principle behind working of RFID tags i.e. identification of products on which RFID tag is put, without requiring any line of sight transponder and RFID reader. In a RFID module the basic transponder has all the information, the transponder is also called RFID tag, the stored information can be in binary bit or lot of binary bits, for example: personal information, medical information, or any type of information, stored digitally.

On coming in range of the reader the RFID tag i.e. a passive tag communicated with the transceiver. Since the passive RFID tags are powerless they take power from

Electromagnetic field coming from RFID reader. Usually the most important component of any tag is the chip or microchip. When tag reaches the reader's RF field, it draws so much power that it can easily access internal memory, and transfer the information stored.

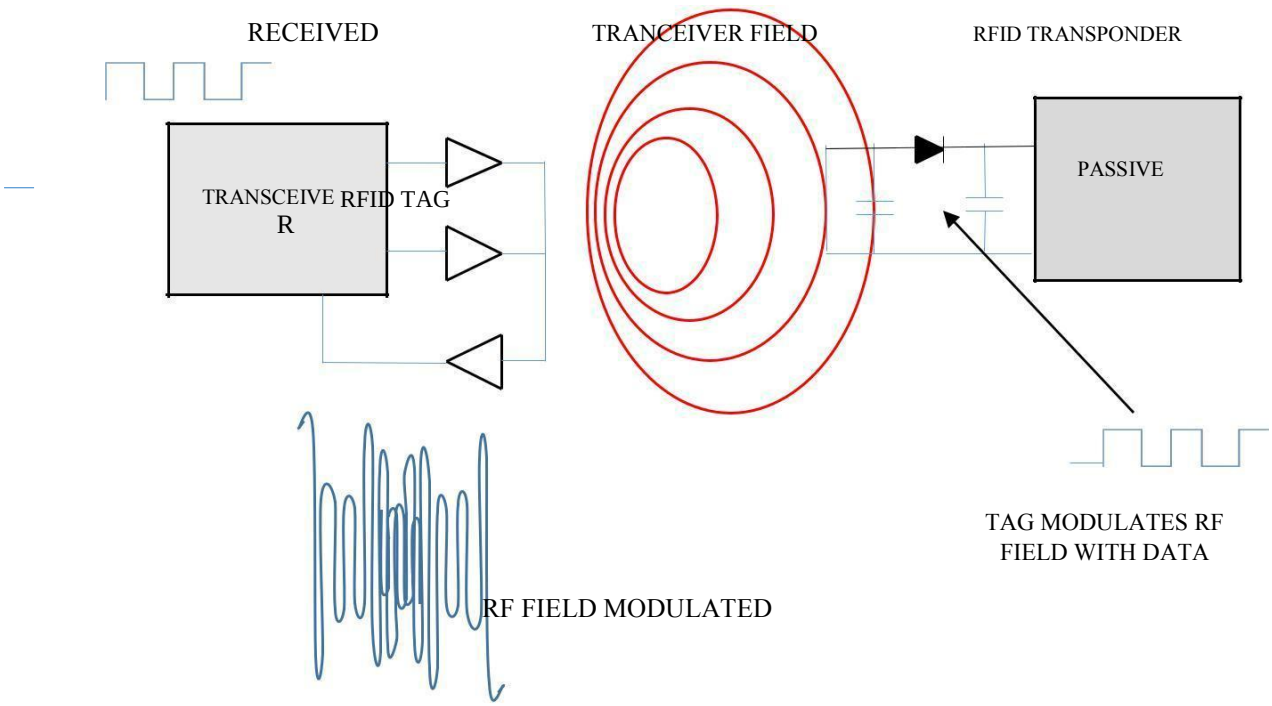

Fig 3.11 Operation of RFID

As soon as transponder gains power the voltage at antenna of tranceiver drops in value due to RF field. This helps the tag to communicate with reader and transfer the stored information.

The tag is capable of controlling the quantity of gained power from RF field, which

enables it to modulate the amount of voltage got at tranceiver accordingvto the pattern of the bits it wants to transfer in form of information.

To carry out such methodology and complete this project a step by step procedure or plan is followed which helped to setup a methodology for proper working of this project.

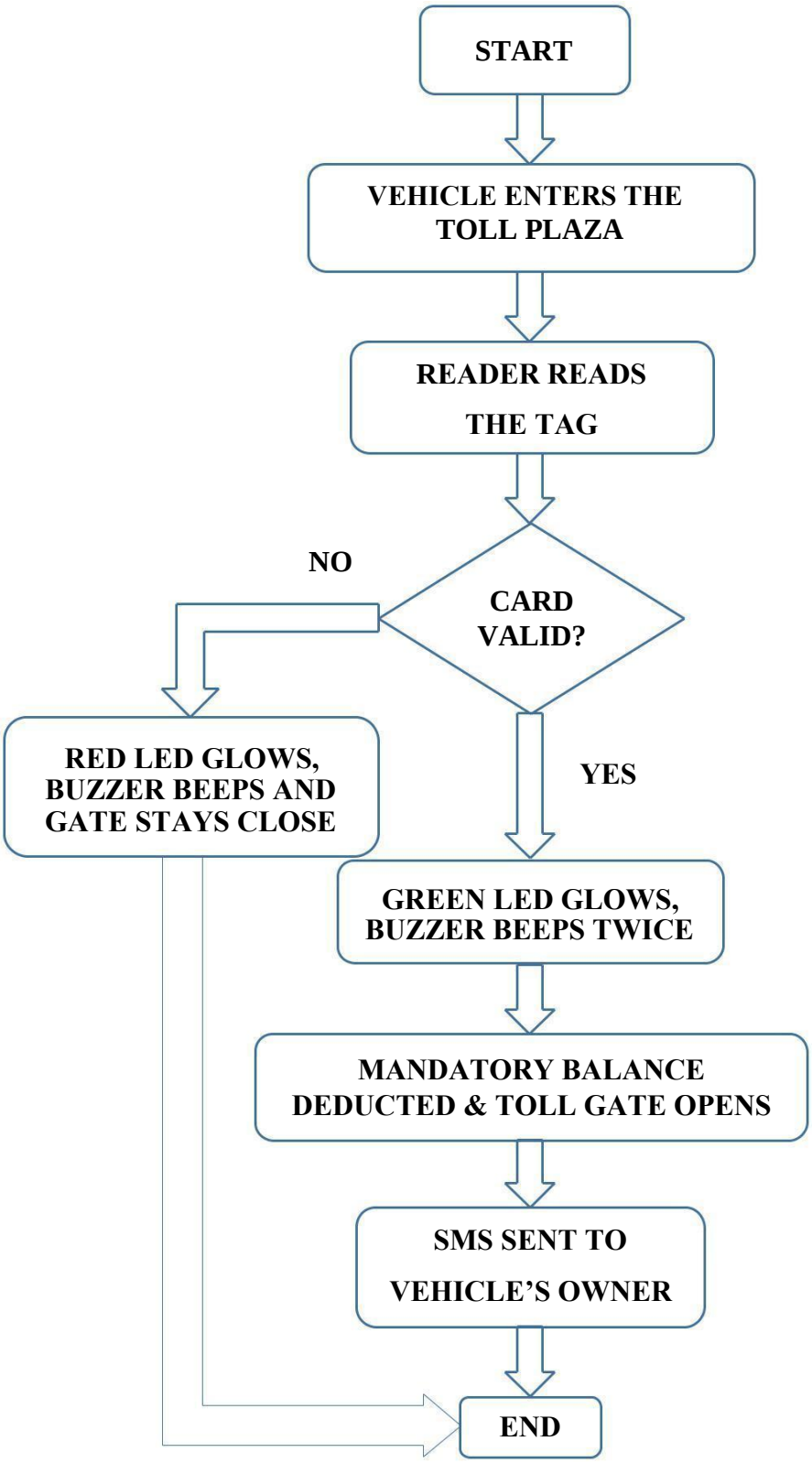

Fig 3.12 Flow chart of Project's Procedure

These steps were followed by developing code for the whole project in section, the below mentioned codes are sequentially presented as developed, to fully make the ATCS model working.

1. Code for Reading RFID Tag: This is basic code common foe reading all RFID tags.

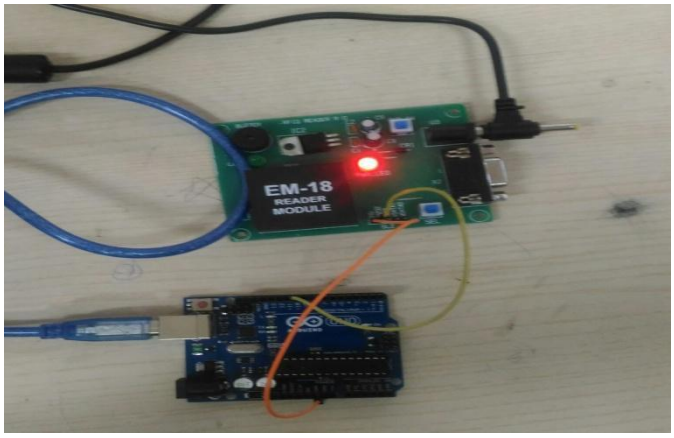

Fig 3.13 RFID Reader Interfaced with ARDUINO

```
#include <SoftwareSerial.h>
SoftwareSerial RFID(10, 11); // RX and TX
char unit;
int index =0;
String tag;
void setup()
{
Serial.begin(9600); // start serial to PC
Serial.println("Serial Ready");
RFID.begin(9600); // start serial to RFID reader
Serial.println("RFID Ready");
}
void loop()
{
while (RFID.available() > 0)
{
unit = RFID.read();
index++;
tag = tag + unit;
```

```
if (index == 12)
{
Serial.print("Tag    ID:
");  Serial.println(tag);
index = 0;
tag = "";
}}}
```

2. Code for working of GSM Module: This code was developed to send SMS through the GSM module using Arduino.

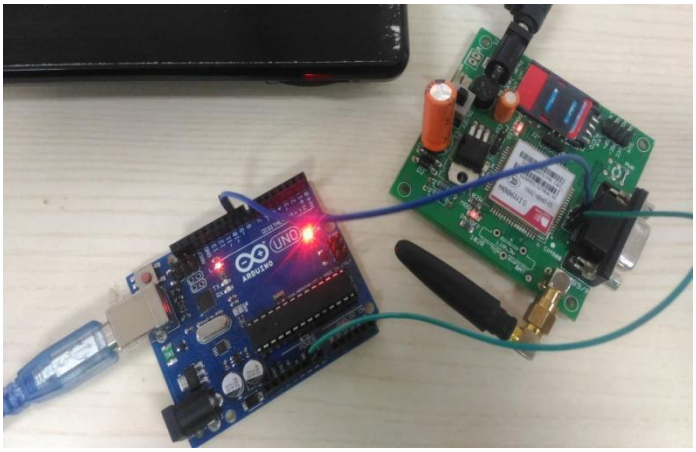

Fig 3.14 ARDUINO, GSM Module Interfacing

```
#include <SoftwareSerial.h>
SoftwareSerial mySerial(9, 3);
void setup()
{
  mySerial.begin(9600); // Setting the baud rate of GSM Module
  Serial.begin(9600); // Setting the baud rate of Serial Monitor (Arduino)
  delay(100);
}

void loop()
{
  if (Serial.available()>0)
   switch(Serial.read())
```

```
{
    case 's':
      SendMessage();
 break;
    case 'r':
   RecieveMessage();
      break;
  }

 if (mySerial.available()>0)
   Serial.write(mySerial.read());
}

 void SendMessage()
{
  mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode
  delay(1000); // Delay of 1000 milli seconds or 1 second
  mySerial.println("AT+CMGS=\"+919408555676\"\r"); // Replace x with mobile
number
  delay(1000);
  mySerial.println("Toll is collected");// The SMS text you want to send
  delay(100);
   mySerial.println((char)26);// ASCII code of CTRL+Z
  delay(1000);
}

 void RecieveMessage()
{
  mySerial.println("AT+CNMI=2,2,0,0,0"); // AT Command to recieve a live
  SMS delay(1000);
 }
```

3. Code for making the motor work: This code opens the gate if the card is valid and deduct the money from the card.

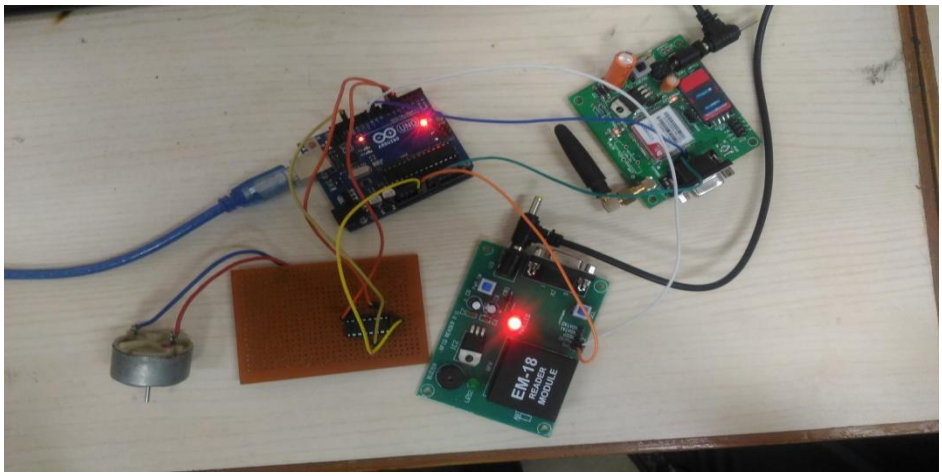

Fig 3.15 ARDUINO, EM-18, Basic Shield & Motor Shield Interfaced

```
#include <SoftwareSerial.h>

SoftwareSerial RFID(10, 11); // RX and TX for RFID Reader
char unit;

int a = 5000; //Jenil's Initial Balance
int y = 4000; //Yash's Initial
Balance int b,x;

int index = 0;

String tag;

void setup()

{

Serial.begin(9600); // start serial to PC
Serial.println("Serial Ready");
RFID.begin(9600); // start serial to RFID
reader Serial.println("RFID Ready");

pinMode(12, OUTPUT); // Initialize digital pin as output (LED Valid Card)
pinMode(9, OUTPUT); // Initialize digital pin as output (LED In-Valid Card)
pinMode(6, OUTPUT); // Initialize digital pin as output (Motor CW)
pinMode(5, OUTPUT); // Initialize digital pin as output (Motor Anti CW)

}
```

```
void loop()
{
digitalWrite(12, HIGH ); // Setting Valid LED OFF
digitalWrite(9, HIGH ); // Setting Invalid LED ON
digitalWrite(6, LOW ); // Setting Motor OFF
digitalWrite(5, LOW ); // Setting Motor OFF while
(RFID.available() > 0)
 {
 unit = RFID.read();
index++;
tag = tag + unit;
if (index == 12)
{
Serial.print("Tag ID: ");
Serial.println(tag);
if (tag == " 10004BB3648C")
{
 b = a-
50; a= b;
Serial.println(tag + ", VALID CARD" + "\n " + "HI JENIL PANDYA"
); digitalWrite(12, LOW ); // set the LED on
delay(1000);
digitalWrite(6, HIGH ); // start the CW motion of motor
delay(300);
digitalWrite(5, LOW ); // keep the Anti CW motion at halt
delay(300);
digitalWrite(6, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(5, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(6, LOW ); // keep the CW motion at halt
delay(550);
digitalWrite(5, HIGH ); // start the Anti CW motion of motor
delay(550);
```

```
Serial.print( "Your Current Balance Is:  " );
Serial.println(b);
}
delay(2000);
digitalWrite(5, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(6, LOW ); // keep the CW motion at halt
delay(550);
digitalWrite(5, HIGH ); // start the Anti CW motion of motor
delay(550);
Serial.print( "Your Current Balance Is:  " );
Serial.println(b);
}
else if (tag == " 10004BA8C93A")
{
x = y-
100; y= x;
Serial.println(tag + ", VALID CARD" + "\n " + "HI YASH SHAH" );
digitalWrite(12, LOW ); // set the LED on
delay(1000);
digitalWrite(6, HIGH ); // start the CW motion of motor
delay(300);
digitalWrite(5, LOW ); // keep the Anti CW motion at halt
delay(300);
digitalWrite(6, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(5, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(6, LOW ); // keep the CW motion at halt
delay(550);
digitalWrite(5, HIGH ); // start the Anti CW motion of motor
delay(550);
Serial.print( "Your Current Balance Is:  " );
Serial.println(x);
}
```

```
else
{
Serial.println(tag + ", INVALID CARD");
digitalWrite(9, LOW ); // set the LED on
delay(2000);
digitalWrite(6, LOW );  // set the LED on
}
index = 0; // Sets the index to 0, so that tag does not
repeat itself tag = ""; // Gives a new input every time
}}}
```

4. Final Code: This is the final code which does all the above jobs as well as sends the SMS to the owner of the car regarding the deduction of money, current balance and the toll plaza crossed.

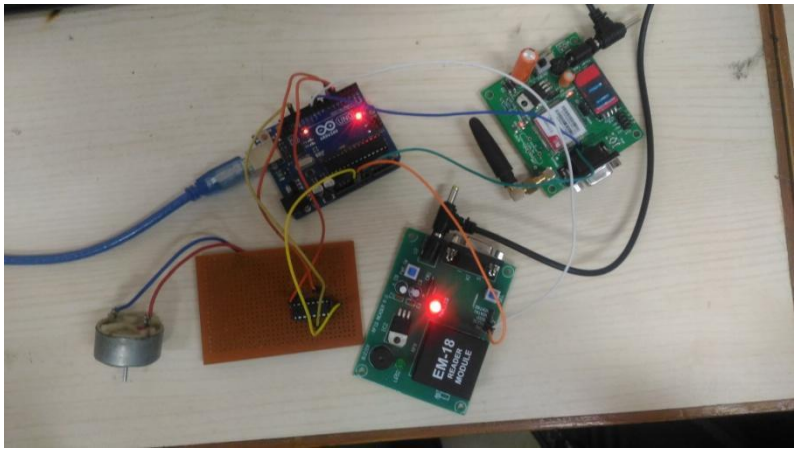

Fig 3.16 ARDUINO, EM-18, Basic Shield, Motor Shield & GSM Module Interfaced

```
#include <SoftwareSerial.h>
SoftwareSerial RFID(10, 11); // RX and TX for RFID Reader
SoftwareSerial mySerial(9, 3); // RX and TX for GSM Module
char unit;
int a = 5000;
int y = 4000;
int b,x;
int index =0;
String tag;
void setup()
{
mySerial.begin(9600); // start serial to GSM
Serial.begin(9600);
delay(100);// start serial to PC
Serial.println("Serial Ready");
RFID.begin(9600); // start serial to RFID
reader Serial.println("RFID Ready");
pinMode(12, OUTPUT); // Initialize digital pin as output (LED Valid Card)
pinMode(4, OUTPUT); // Initialize digital pin as output (LED Valid Card)
pinMode(6, OUTPUT); // Initialize digital pin as output (Motor CW)
pinMode(5, OUTPUT); // Initialize digital pin as output (Motor Anti CW)
}
void loop()
{
digitalWrite(12, HIGH ); // Setting Valid LED OFF
digitalWrite(4, HIGH ); // Setting Invalid LED ON
digitalWrite(6, LOW ); // Setting Motor OFF
digitalWrite(5, LOW ); // Setting Motor OFF while
(RFID.available() > 0)
{
unit = RFID.read();
index++;
tag = tag + unit;
```

```
if (index == 12)
{
Serial.print("Tag ID: ");
Serial.println(tag);
if (tag == "10004BB3648C")
{
b = a-
50; a= b;
Serial.println(tag + ", VALID CARD" + "\n " + "HI JENIL PANDYA"
); digitalWrite(12, LOW ); // set the LED on
delay(1000);
digitalWrite(6, HIGH ); // start the CW motion of motor
delay(80);
digitalWrite(5, LOW ); // keep the Anti CW motion at halt
delay(80);
digitalWrite(6, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(5, LOW ); // brakes applied to motor
delay(2000);
digitalWrite(6, LOW ); // keep the CW motion at halt
delay(160);
digitalWrite(5, HIGH ); // start the Anti CW motion of motor
delay(160);
Serial.print( "Your Current Balance Is:  " );
Serial.println(b);
Serial.println("License No. GJ 11 AE 0840");
mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode
delay(1000); // Delay of 1000 milli seconds or 1 second
mySerial.println("AT+CMGS=\"+919408555676\"\r"); // Messages to
this number delay(1000);
String aa="Hello Jenil. Rs 50 deducted for your car ( GJ 11 AE 0840) at Nirma
Toll Plaza & your balance is Rs "; // The SMS text you want to send
String aaa=aa+a; // String addition or
concatenation mySerial.println(aaa); //string output
```

```
delay(1000);

mySerial.println((char)26);//ASCII code of CTRL+Z
delay(1000);

}

else if (tag == "10004BA8C93A")

{
x = y-
100; y= x;
Serial.println(tag + ", VALID CARD" + "\n " + "HI YASH SHAH" );
digitalWrite(12, LOW ); // set the LED on

delay(1000);

digitalWrite(6, HIGH ); // start the CW motion of motor
delay(80);

digitalWrite(5, LOW ); // keep the Anti CW motion at halt
delay(80);

digitalWrite(6, LOW ); // brakes applied to motor
delay(2000);

digitalWrite(5, LOW ); // brakes applied to motor
delay(2000);

digitalWrite(6, LOW ); // keep the CW motion at halt
delay(160);

digitalWrite(5, HIGH ); // start the Anti CW motion of motor
delay(160);

Serial.print( "Your Current Balance Is:  " );

Serial.println(x);

Serial.println("License No. GJ 01 JV 9071");
mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode
delay(1000); // Delay of 1000 milli seconds or 1 second

mySerial.println("AT+CMGS=\"+917778875077\"\r"); //Messages to this
number delay(1000);

String bb="Hello Yash. Rs 100 deducted for your car ( GJ 01 JV 9071 ) at Nirma
Toll Plaza & your balance is Rs "; // The SMS text you want to send

String bbb=bb+y; //string addition or concatenation

mySerial.println(bbb); //string output

delay(1000);
```

```
mySerial.println(bbb); //string output
delay(1000);
mySerial.println((char)26);// ASCII code of CTRL+Z
delay(1000);
}
else
{
Serial.println(tag + ", INVALID CARD");
digitalWrite(4, LOW ); // set the LED on
delay(2000);
} index = 0; // Sets the index to 0, so that tag does not repeat itself
tag = " "; // Gives a new input every time
}}}
```

This was all about the methodology used for this project, Embedded C Programming, RFID working and connecting small module to make a big module to make ATCS model work.

Chapter 4

RESULTS AND DISCUSSIONS

4.1 Relevant Codes and Results

The whole project was based on Arduino and hence the coding and de-bugging was done on the Arduino 1.6.8 – 1.6.9v of the software of the same name. Instead of going or a single program that is debugged all the time, the project was done by framing out various different codes that gave outputs based on their work and interfacing of the Arduino board with different components of the project.

4.1.1 RFID Read

The RFID Reader i.e. the EM-18 RFID tag reader was interfaced first with the Arduino ATMega328 board so as to read the card or the tag attached to the vehicle. This gave the output as desired showing the tag ID saved into the small memory of the tag.

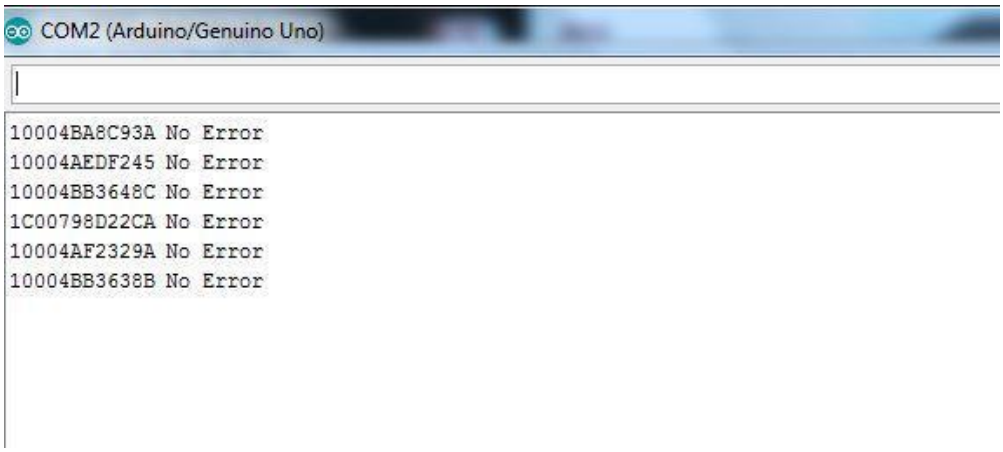

Fig 4.1 RFID Reading Output

4.1.2 Sending SMS using GSM Module

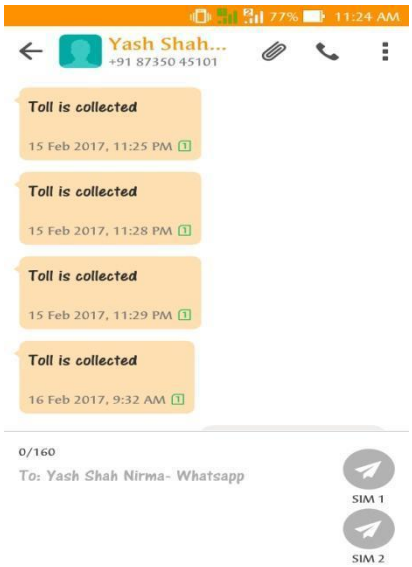

Fig 4.2 GSM Ouput SMS Sent

4.1.3 Money Deduction

A simple and handy program was written to state as to how much balance the IDs are given so that when detected a certain decided amount is to be deducted each time. The money deduction program worked perfectly.

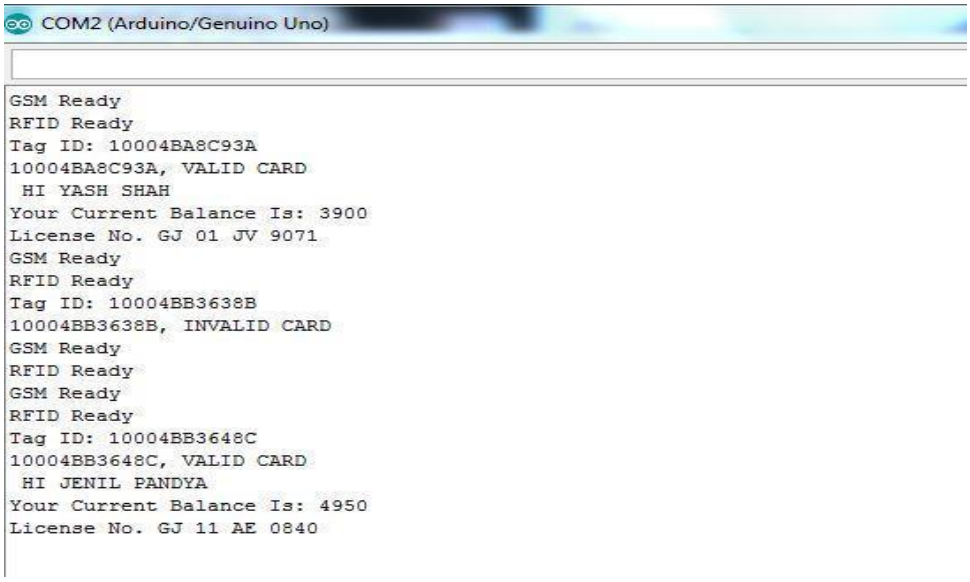

Fig 4.3 Money Deduction Output

4.1.4 GSM – Arduino Interfacing

The next step was to interface the GSM Sim900 Module and the Arduino board. The module helps in sending the user of a RFID Tag a message indicating the amount deducted from his/her account and also displaying the amount left and other details.

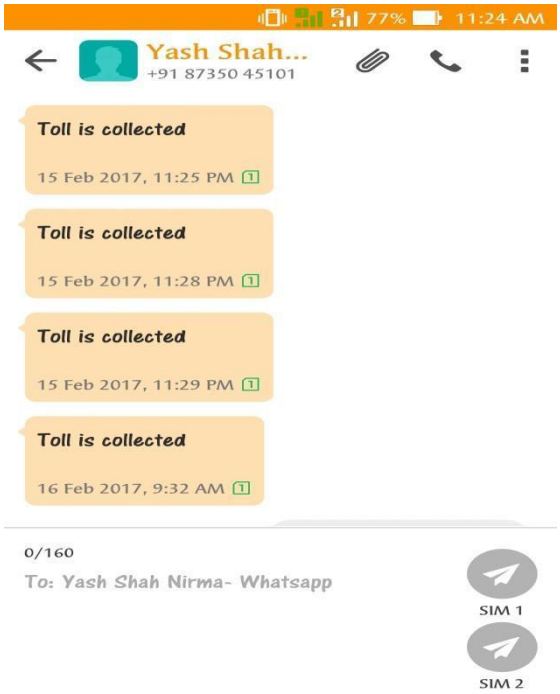

Fig 4.4 Received Test Message

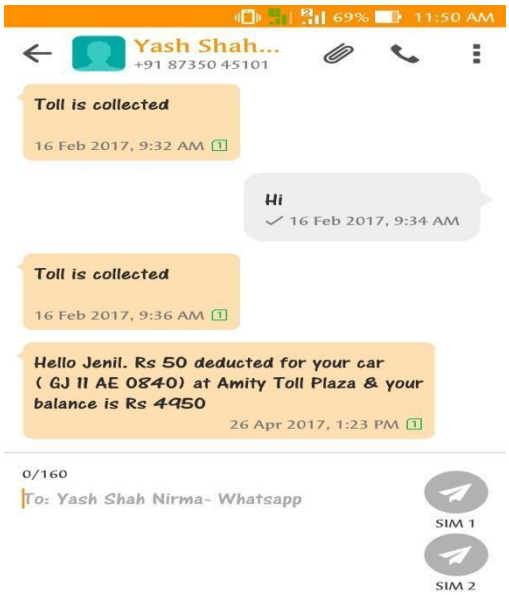

Fig 4.5 Money Deduction Message

4.1.5 Setting up Motor, LEDs and Buzzer

The next step was to set up the final components of the project that comprises stepper motor, LEDs and a buzzer. The whole setup was to make it look like a toll booth that allows the vehicle to pass when the LEDs turn green and the motor lifts up the barrier with buzzer beeping with the LEDs.

4.1.6 Results

The toll collection system works as proposed; when a vehicle approaches the toll booth and reaches the EM 18 card reader, it reads the card as it matches the same frequency range i.e. 125 KHz. The data thus stored in the card i.e. the unique ID is sent to the Arduino where it processes as to what needs to be done. If the program contains the ID read from the card, it gives an output that it is valid/ correct and hence forwards the command to the LEDs and Buzzer to work accordingly i.e. to blink twice and buzz simultaneously. After this is done the Arduino commands the motor to start for a certain time so as to open the toll gate or barrier in a way that the vehicle containing the card could pass easily. After a delay of nearly 5 seconds the toll gate automatically closes so that no other vehicle could pass.

The Arduino further commands the GSM Sim900 Module to forward a message meant only for that registered user suggesting that a certain amount has been deducted and notifying that this much amount is left in their balance for toll. This ends the procedure. If the card is read and the Arduino rejects the card displaying it as invalid/incorrect, a command is sent to the LEDs and the buzzer to beep for a long duration and hence not opening the gate.

4.2 Discussion

In this project, a system that collects toll automatically using electronic devices has been proposed. Since, our conventional methods for collecting toll fees or roadways taxes are not very efficient and reliable so there was a need for a new advanced system that eliminates most of the drawbacks of the current system in use.

This technology is not very familiar in most of the countries. Since, it requires a highly efficient infrastructure and needs to be well maintained. Also, the cost of providing these RFID cards to each and every vehicle is not very cost efficient. As in present modern era, the number of vehicles has increased rapidly. But, apart from these shortcomings of the system there are no major flaws that may give us a thought to drop the idea or lower our interests in such systems. The perks and gains from this system are far beyond their flaws.

The benefits of the proposed system are:

1. Time Saving: This is one of the major benefits of ETC systems. Since, user is not required to stop and pay the cash. So, it saves a lot of time.
2. Reduced Congestion/ traffic: ETC toll plazas play a major role in reducing the traffic. The manual systems make the vehicles stop and pay the cash to the collector. Thus, making a long trail of vehicles waiting for their chance to pay.

   But in ETC plazas, the users don’t stop to pay the cash unless there is a problem regarding their identity. It drives directly through the plaza and avoid a long line of vehicles.
3. Saves Fuel: In manual systems, when vehicles wait for their chance to come they don’t shut the engines off. This makes the vehicle to consume more fuel. Whereas in ETC system, the vehicle doesn’t stop and passes directly without wasting any fuel.
4. Cash Handling enhanced: In ETC systems, there is no use of cash. So, all the problems related with cash handling are eliminated.
5. Database enhanced: In manual systems, it is very difficult to maintain the database such as the number of vehicles passing the toll plaza in a single day, the amount of cash collected in one day etc.

   But, this maintenance problem is eliminated in ETC systems. Since, all the work is computerized and automated.

6. Capacity of the Lane increased: now a day, the number of lanes in a road is increasing. We have 8 lane highways and 16 lane highways. This requirement was felt due to congestion. But, ETC systems can eliminate this problem which can increase the capacity of the lane.
7. Reduced number of accidents: Since, there is no congestion around the toll plaza; the chances of accidents are also reduced due to these systems.
8. Reduced human error: In ETC systems, there is no need of a collector or any human to operate. Thus, the chances of human errors are also reduced. The database may face faults due to human error which is reduced in these systems.
9. Reduced corruption: Since, there is lesser manual interruption in ETC systems; the chances of corruption are also reduced.
10. Environmental aspects: Apart from all the above benefits, ETC systems also have few points favorable to environment. Fuel consumption has been discussed. Other than this, there is one more aid of this system i.e. the use of SMS service as payment receipt instead of paper. This saves a very large amount of paper.

All the above benefits prove that ETC systems have a very great use in future and will be accepted by the national highway authorities.

Chapter 5

CONCLUSION

5.1 Conclusion

At the ending of this report, the conclusion that came out is that, to implement modern system of toll collection i.e. ATCS embedded systems were used and a new technique RFID came into light. The RFID safety is a key important thing about this project. Strong and verified designs were kept in the project, and good, reliable RFID technique was used for authentication and authorization of vehicles. This project is reliable and easy way to collect toll comparing to manual one. ATCS proposed will significantly improve travelling through highways by reducing wastage of time and fuel. This system can easily collect toll from people without even making them stay at the toll booths. This is achieved by using wireless technique of data transfer from vehicle to microcontroller and from microcontroller to vehicle.

This system bill definitely help both the ends i.e. toll authorities and the people in form of cost, time, increased capacity and better convenience and security. Due to implementation of this technique at all toll plazas of India, the problem of long ques and need for human intervention will be reduced. In ATCS basically a RFID tag is the gate pass for the vehicle if its valid one can go through if its invalid the owner has to wait. This idea of toll collection also reduces fuel consumption and pollution.

So, it is clear from this report that RFID base toll collection system is not only better than manual system but is the best automated toll collection system technique by far because it also helps the owner of the vehicle to know the current location of his/her vehicle through the SMS received from toll plaza after successful transaction, carrying information about the toll plaza where transaction took place

Chapter 6

SUMMARY

The overall summary of this report and project on Automated Toll Collection System using RFID is that, by all the literature reviews discussions and brain boosting it is clear the ATCS is being used in lot of parts of the word with different principles like, microwave, image processing, GNSS, infrared technique, barcode and many more, but the most successful ATCS have been in countries with RFID based toll collection, because of lot of reasons which include low cost, easily available, secure, reliable and most important is fastest of all other techniques. This is because the RFID ATCS proposed in this project works in following way.

As car approaches the toll booth. The RFID reader at the toll will read the RFID card embedded on the vehicle. Then, it will pass the information stored in the card to the Arduino ATmega 328.The Arduino will track information and will check whether the card is valid or not. If the card is valid, the owner's name will be displayed on the screen.

A certain amount will be deducted from the owner's account. Here, we have fixed the amount for different users. For Jenil, the amount is Rs 50 and for Yash it is Rs100. Along with this, a green LED light will glow twice with the buzzer doing the same. After this, a command will be send to the motor to open the gate and vehicle will pass.

Now, GSM module will send message to the owner informing him about amount deducted from their tag. Here, message will be: 'Rs 50/100 deducted'. If the card is not valid, the screen will show 'invalid user' and no amount would be deducted. Along with this, a red LED light will glow with the buzzer making noise continuously for 2 seconds. Thus, no command will be send to the motor. And the toll gate will remain close.

There have been a lot of reasons for making this project which found during our surveys and research, there are some facts that will make it clear how important it is to install RFID based ATCS in INDIA.

The financial and transportation report of 2015-2016 states that:

Approx. 21,08600 vehicle go through highways every year

At least 50 seconds are wasted per day for one vehicle to pay toll
Meaning a vehicle wait for 6 hours per year at toll plazas

Through this technique time for waiting of a vehicle e day is reduced to 11 sec.
Which is yearly 1.2 hours, leading to save almost saving 5 hours a year as well

as lot of fuel money and time.

The summary of this project also tells that most important part of ATCS is RFID, as it is only technology which makes this project flexible, feasible & better than others as it can be further used in animal tracking, human implants, speed tracking etc. Use of different ATCS tech. is shown in table mentioned below:

Table 4. Comparison of Different ATCS Techniques

ATCS Type	Cost	Places in use	Specifications
Infrared	1000/- unit 2lac/- reader	Malaysia	Easily used in lot of applications
Active Microwave	2000/- unit 5lac/- reader	Japan	High bandwidth, high data rate
Passive Microwave	2000/- unit 2lac/- reader	Europe	Very simple on board unit
Passive RFID	100/- tag 2lac/- reader	US, Georgia	Tamper resistant tags, small, light, cheap, almost unlimited life
Active RFID	1000/- tag 2lac/- reader	Florida	On board transmitter, expensive, high range
GNSS/CNN	2000/- tag 2lac/- reader	Germany	Too sophisticated, no toll plazas so violation very easy in INDIA

As this table shows that RFID based ATCS is best still there are some problems faced by this system which can be solved in future like low frequency of tag lowers the data transfer rate, making transmission of data really slow, duplicity of RFID tag is quite easy, and many more.

Chapter 7

REFERENCES

[1]. Khadijah K, Dr. Widad I (2005), *Electronic Toll Collection System using Passive RFID Technology*, International Journal of Science and Research (IJSR), I: 70, Vol. No. 4, Issue 2.

[2]. Priyanka C, Govinda P, Poonam B (2013*), Image Processing Based Automatic Toll Booth in Indian Conditions*, International Journal of Computer Science and Information Technology (IJCSIT), I:410, III:411-412, Vol. No.5.

[3]. Pranoti S, Poonam M, Kirti D, Jayshree D (2013), *Automated Toll Collection System Using RFID*, International Journal of Advanced Research in Electrical Electronics & Instrumentation Engineering(IJAREEIE), II: 62, Vol No.3, Issue.

[4]. Rakhi K, Anand P, Akshay M, Rohan K (2014), *RFID Based Toll Collection System*, International Journal of Electrical and Electronics Research (IJEER), 3: 2582 Vol No.2 Issue -2.

[5]. Figure 1.1: https://icons8.com/web-app/2354/rfid-tag

[6]. Figure1.2: http://www.francetravelplanner.com/trans/car/tolls.html

[7]. Figure2.1: http://www.rfid.com/internalworking

[8]. Figure2.3: http://www.tollbooths.com/manual

[9]. Microcontrollers & Embedded Systems by Muhammad Ali Mazidi, Janice Gillispie Mazidi, Prentice-Hall.